AI ASSISTED CODING ASS – 14.1

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B-27

Task 1 – Portfolio Website Design

You are building a personal portfolio website to showcase your work.

Requirements:

Create sections for About Me, Projects, and Contact.

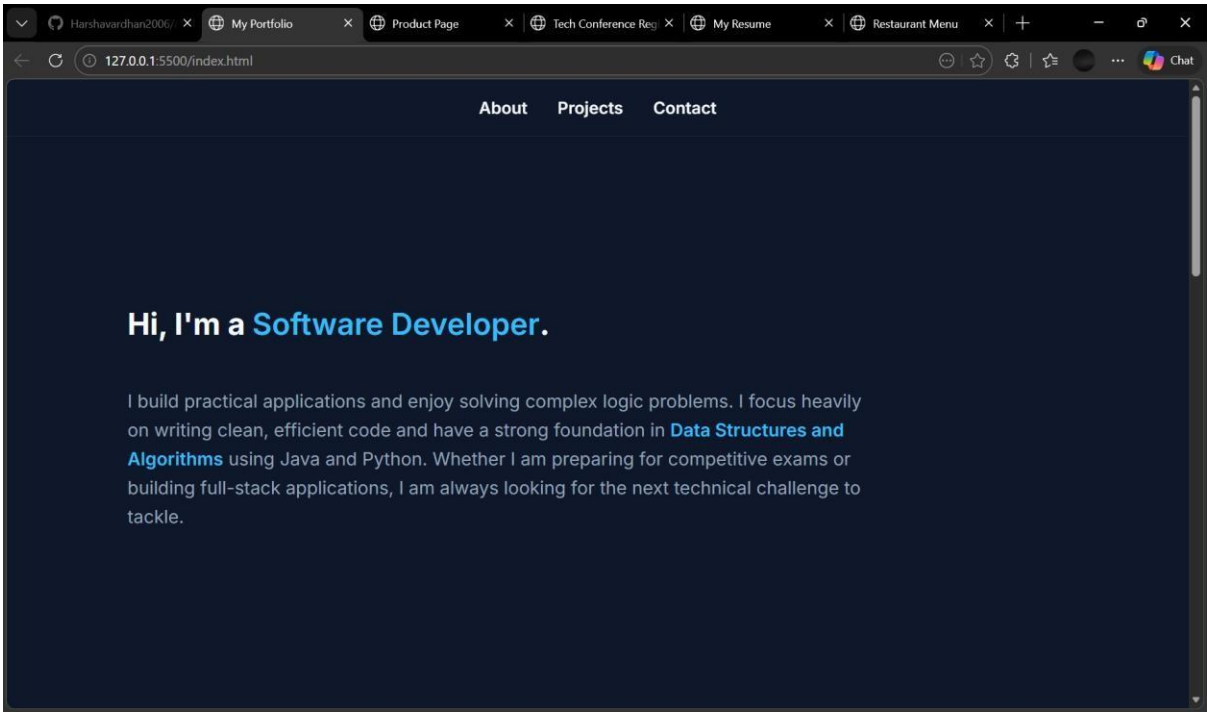
Use AI to:

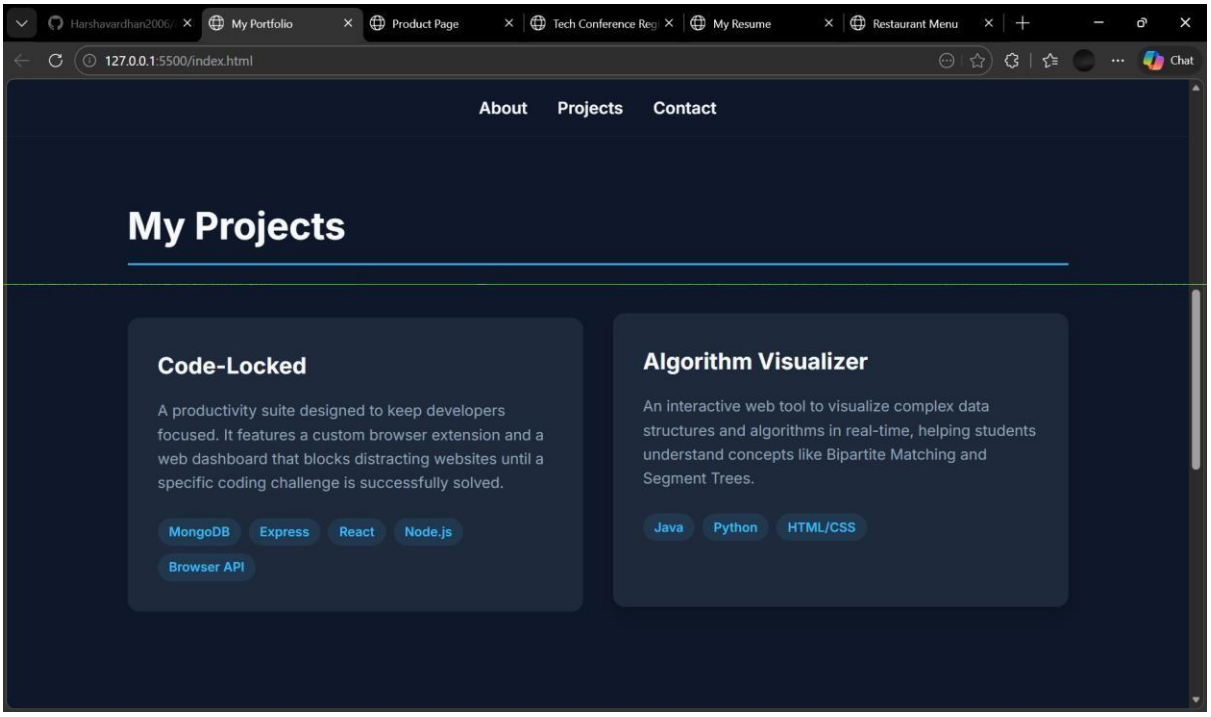
- Suggest color palettes and typography.
- Create a responsive layout with Grid/Flexbox.
- Add smooth scrolling navigation.

CODE:

```html
<!-- create a portfolio website to showcase my work and skills
 create sections for AboutMe, Projects, and Contact, suggest color palettes and typography
 and create a responsive layout with grid/Flexbox and add smooth scrolling navigation -->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Portfolio</title>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;600;700&display=swap"
    rel="stylesheet">
    <style>
        /* Color Palette & Typography */
        :root {
            --bg-color: #0f172a;      /* Deep Slate */
            --surface-color: #1e293b; /* Lighter Slate for cards */
            --accent-color: #38bdf8;  /* Cyan/Light Blue */
            --text-main: #f8fafc;     /* Off-White */
            --text-muted: #94a3b8;    /* Gray */
            --font-main: 'Inter', sans-serif;
        }

        /* Base Styles & Smooth Scrolling */
        html {
            scroll-behavior: smooth;
        }

        body {
            margin: 0;
            padding: 0;
            font-family: var(--font-main);
            background-color: var(--bg-color);
            color: var(--text-main);
            line-height: 1.6;
        }

        a {
            text-decoration: none;
            color: var(--accent-color);
```

Output:


Harshavardhan2006/
My Portfolio
Product Page
Tech Conference Reg
My Resume
Restaurant Menu
127.0.0.1:5500/index.html
Chat
About
Projects
Contact
Hi, I'm a Software Developer.
I build practical applications and enjoy solving complex logic problems. I focus heavily on writing clean, efficient code and have a strong foundation in Data Structures and Algorithms using Java and Python. Whether I am preparing for competitive exams or building full-stack applications, I am always looking for the next technical challenge to tackle.


Harshavardhan2006/
My Portfolio
Product Page
Tech Conference Reg
My Resume
Restaurant Menu
127.0.0.1:5500/index.html
Chat
About
Projects
Contact
My Projects
Code-Locked
A productivity suite designed to keep developers focused. It features a custom browser extension and a web dashboard that blocks distracting websites until a specific coding challenge is successfully solved.
MongoDB
Express
React
Node.js
Browser API
Algorithm Visualizer
An interactive web tool to visualize complex data structures and algorithms in real-time, helping students understand concepts like Bipartite Matching and Segment Trees.
Java
Python
HTML/CSS

Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

Display product image, title, price, and "Add to Cart" button.

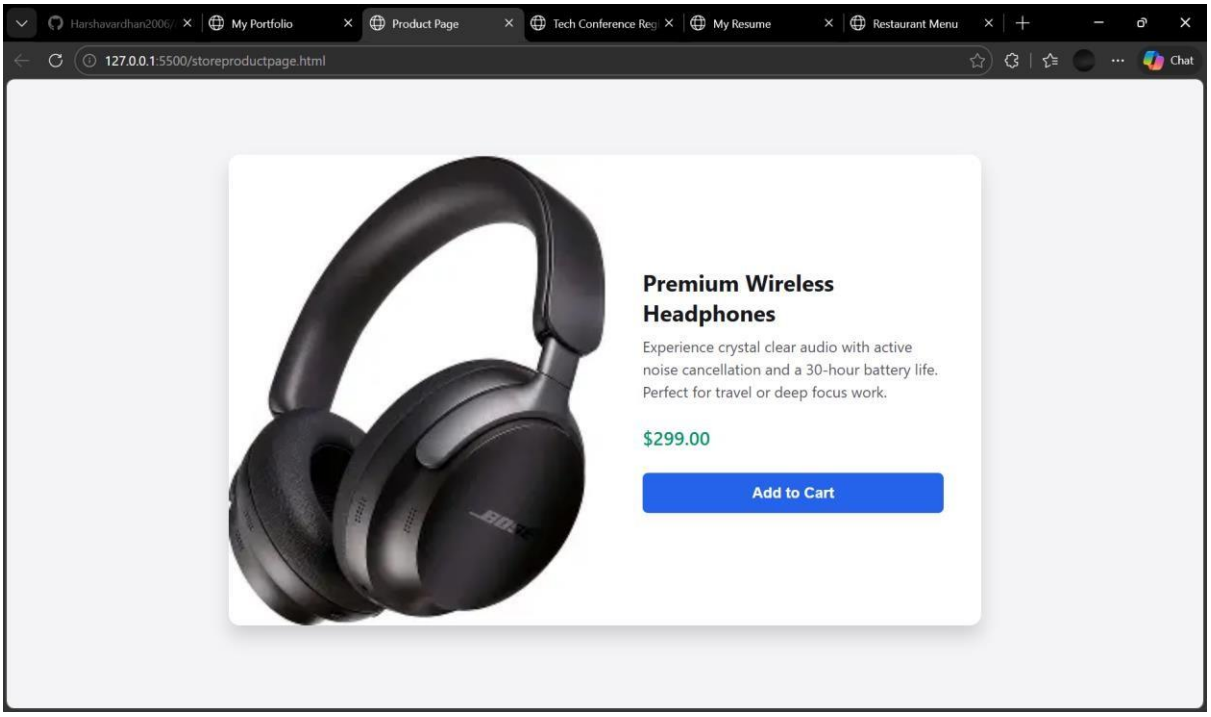
Use AI to:

- Style with BEM methodology.
- Make layout responsive.
- Add hover effects and "Add to Cart" alert.

CODE:

```html
<!-- Create a Store Product Page, display product image, title, price, and "Add to cart" button
 style with BEM methodology, make layout responsive and hover effects and "Add to cart" alert -->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Product Page</title>
    <style>
        /* Base Reset */
        * {
            box-sizing: border-box;
            margin: 0;
            padding: 0;
        }

        body {
            font-family: system-ui, -apple-system, sans-serif;
            background-color: #f4f4f5;
            display: flex;
            justify-content: center;
            align-items: center;
            min-height: 100vh;
            padding: 20px;
        }

        /* BEM Block: product-card */
        .product-card {
            display: flex;
            flex-direction: column;
            background: #ffffff;
            border-radius: 12px;
            overflow: hidden;
            box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
            max-width: 400px;
            width: 100%;
            transition: transform 0.2s ease, box-shadow 0.2s ease;
        }
```

Output:



Task 3 – Event Registration Form

Build an event registration form for a conference.

Requirements:

Collect name, email, phone number, and session selection.

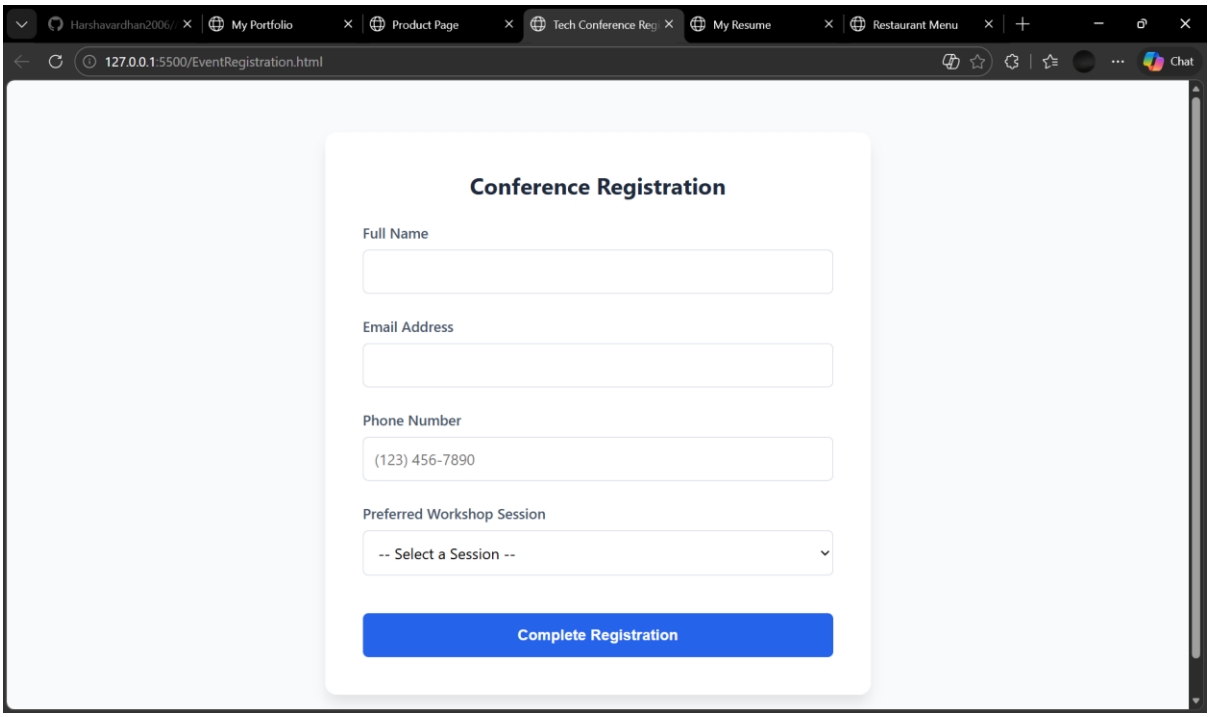
Use AI to:

- Add form validation with JavaScript.
- Make the form accessible with labels and ARIA.
- Style with a professional look

CODE:

```html
<!-- Create Event Registration Form which collects name, email, phone number, and session selection
add form validation with JS and make the form sccessible with labels and ARIA and style with a professional look -->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Tech Conference Registration</title>
    <style>
        /* Professional Styling */
        :root {
            --primary-color: #2563eb;
            --primary-hover: #1d4ed8;
            --bg-color: #f8fafc;
            --card-bg: #ffffff;
            --text-main: #1e293b;
            --text-label: #475569;
            --border-color: #cbd5e1;
            --error-color: #dc2626;
            --success-color: #16a34a;
        }

        body {
            font-family: system-ui, -apple-system, sans-serif;
            background-color: var(--bg-color);
            color: var(--text-main);
            display: flex;
            justify-content: center;
            align-items: center;
            min-height: 100vh;
            margin: 0;
            padding: 20px;
        }

        .registration-container {
            background-color: var(--card-bg);
            padding: 2.5rem;
            border-radius: 12px;
            box-shadow: 0 10px 15px -3px rgba(0, 0, 0, 0.1);
            width: 100%;
```

Output:



Task 4 – Resume Website

Build a one-page online resume.

Requirements:

Sections: Profile Summary, Skills, Education, Experience.

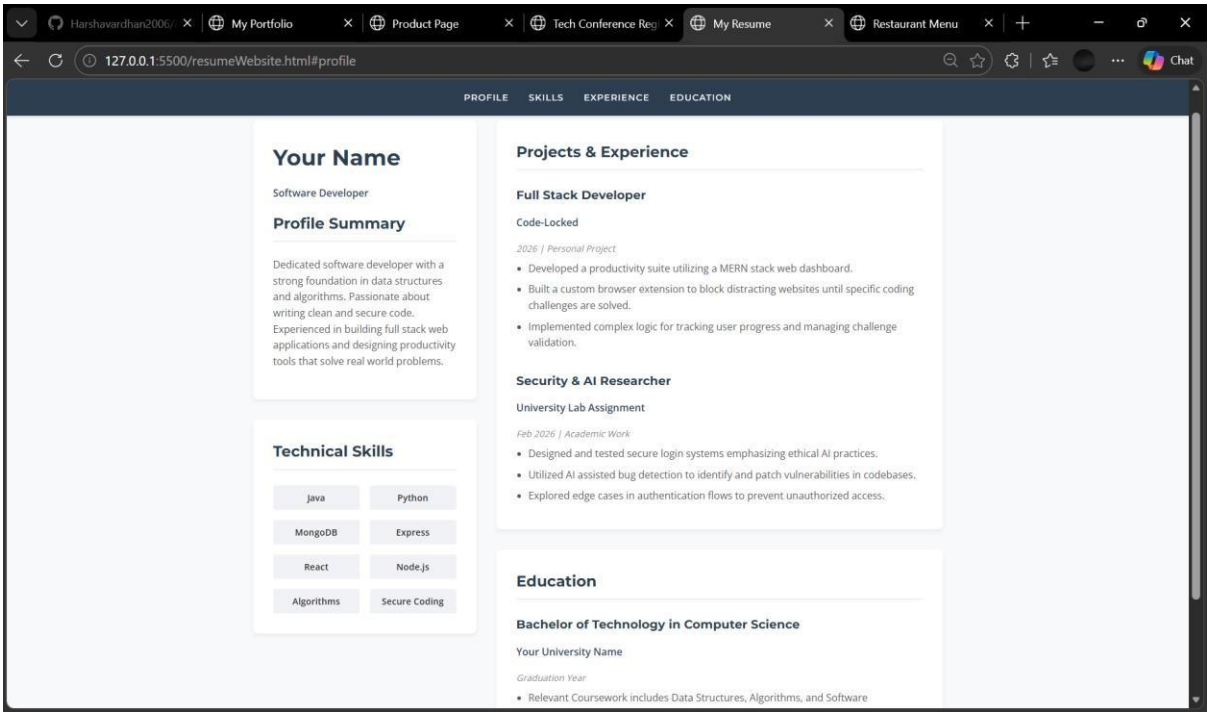
Use AI to:

- Propose professional typography styles.
- Align sections using CSS Grid.
- Add smooth scroll navigation between sections.

CODE:

```html
<!-- Create a Resume Website which conatins the sections profile summary, education, experience, and skills
 propose professional typography styles, align sections using CSS grid and add smooth scroll navigation between sections -->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My Resume</title>
    <link href="https://fonts.googleapis.com/css2?family=Montserrat:wght@500;700&family=Open+Sans:wght@400;600&display=swap" rel=
    <style>
        /* Professional Typography and Colors */
        :root {
            --bg-page: #f8f9fa;
            --bg-card: #ffffff;
            --text-main: #2b2b2b;
            --text-secondary: #5a5a5a;
            --accent-color: #2c3e50;
            --accent-light: #34495e;
            --font-heading: 'Montserrat', sans-serif;
            --font-body: 'Open Sans', sans-serif;
        }

        html {
            scroll-behavior: smooth;
        }

        body {
            margin: 0;
            padding: 0;
            font-family: var(--font-body);
            background-color: var(--bg-page);
            color: var(--text-main);
            line-height: 1.6;
        }

        /* Navigation */
        nav {
            position: fixed;
            top: 0;
            width: 100%;
```

Output:



Task 5 – Restaurant Menu Page

Design a digital menu page for a restaurant.

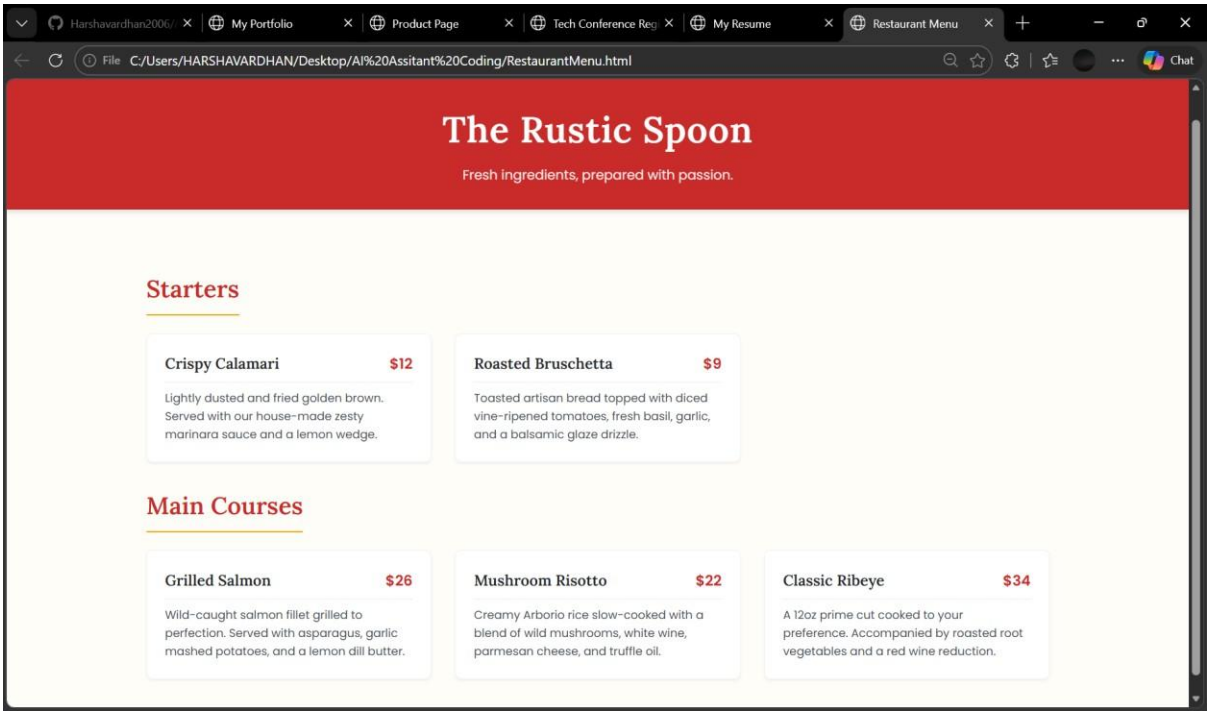
Requirements:

Display menu items with name, description, and price.

Use AI to:

- Suggest a food-friendly color palette.
- Design a responsive card-based layout.
- Add hover effects on menu items.

CODE:

```html
<!-- create a Restaurant Menu which displays menu items with names, descriptions, and prices
 Suggest a food-friendly color palette, design a responsive card-based layout and hover effects on menu items -->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Menu</title>
    <link href="https://fonts.googleapis.com/css2?family=Lora:ital,wght@0,400;0,600;1,400&family=Poppins:wght@400;600&display=swap
    <style>
        /* Food-friendly Color Palette */
        :root {
            --bg-color: #fffdf7;       /* Creamy off-white */
            --card-bg: #ffffff;        /* Pure white for cards */
            --primary-color: #c92a2a; /* Deep appetizing red */
            --accent-color: #f59f00;  /* Warm golden yellow */
            --text-dark: #212529;      /* Deep gray/black */
            --text-muted: #495057;     /* Lighter gray for descriptions */
            --font-heading: 'Lora', serif;
            --font-body: 'Poppins', sans-serif;
        }

        body {
            margin: 0;
            padding: 0;
            font-family: var(--font-body);
            background-color: var(--bg-color);
            color: var(--text-dark);
            line-height: 1.6;
        }

        header {
            text-align: center;
            padding: 4rem 1rem 2rem;
            background-color: var(--primary-color);
            color: #ffffff;
            margin-bottom: 3rem;
            box-shadow: 0 4px 6px rgba(0,0,0,0.1);
        }
```

Output: